An Integrated Conceptualization and Interactive Visualization of Bragg, Laue, Miller, Woolfson, Ewald, and Gibbs

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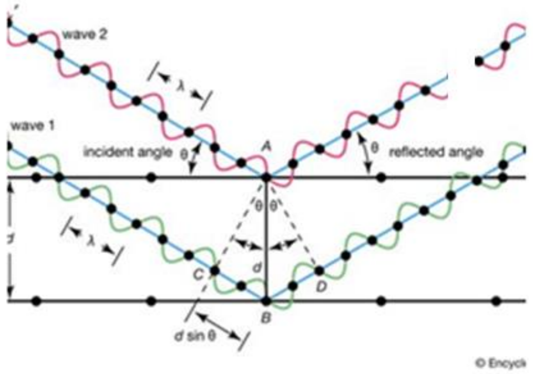
Latin American Crystallographic Association
Sixth Annual Meeting
Montevideo, Uruguay
25 September 2024

Acknowledgments

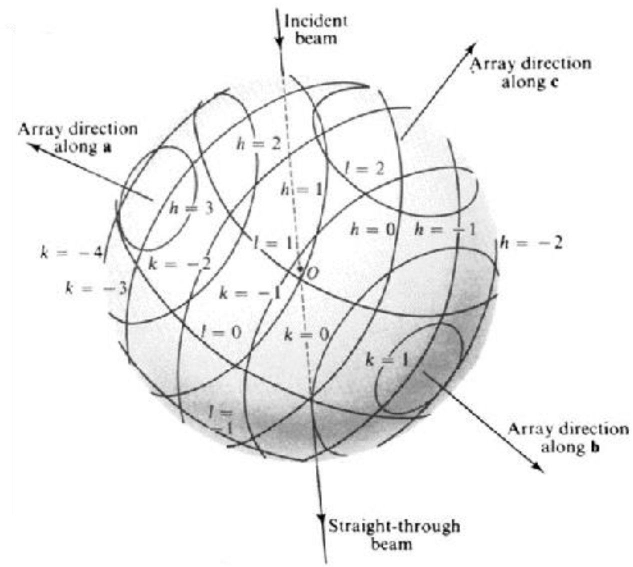
Huge thanks to Leo Suescun, who took the time to patiently teach me about crystallography.

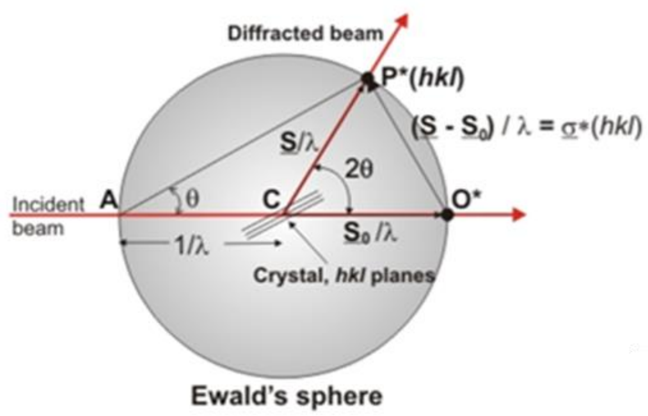
Northfield Minnesota, July 2024; basking in the triumph of a 50 km (hilly!) bicycle ride

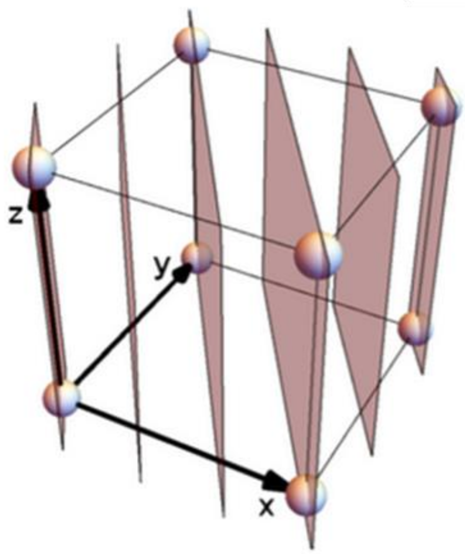
Overview

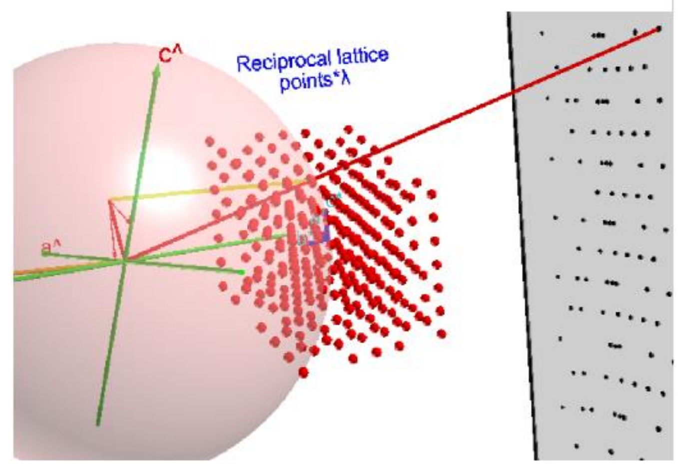


$$\mathbf{a}\cdot\mathbf{s} = h$$
$$\mathbf{b}\cdot\mathbf{s} = k$$
$$\mathbf{c}\cdot\mathbf{s} = l$$



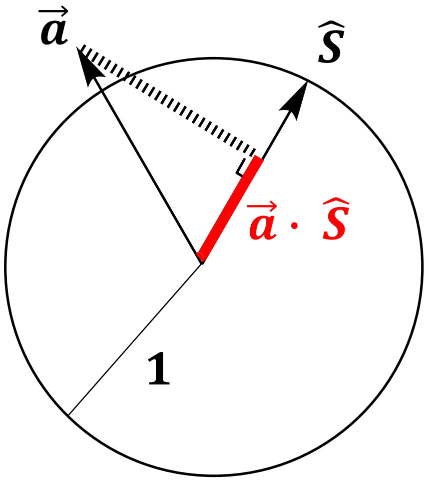






The Basic Idea

The dot product of a vector $\vec{\boldsymbol{a}}$ with a *unit vector* $\hat{\boldsymbol{s}}$ can be visualized as a ***projection****:*

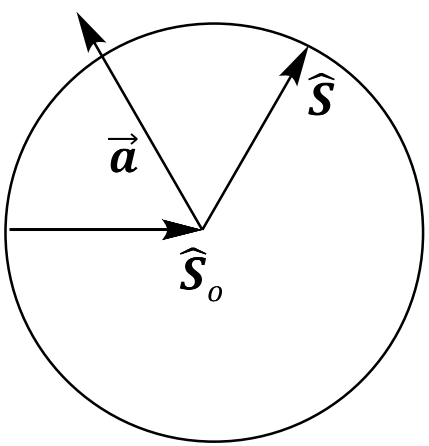


1. Laue

$$\vec{\boldsymbol{a}} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = h\lambda$$
$$\vec{\boldsymbol{b}} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = k\lambda$$
$$\vec{\boldsymbol{c}} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = l\lambda$$

$\widehat{\boldsymbol{S}}$ and $\widehat{\boldsymbol{S}}_o$ are *unit vectors.*

Dotting their difference with each crystallographic axes must give a multiple of the X-ray wavelength.



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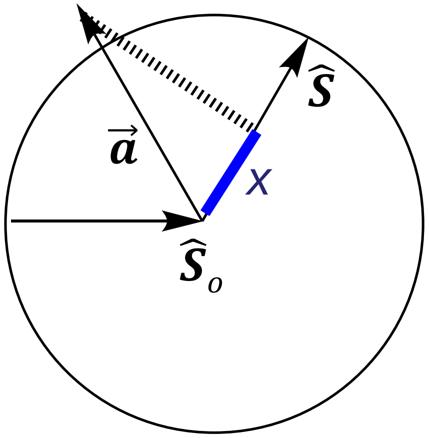
$x = \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}}$

$\widehat{\boldsymbol{S}}$ and $\widehat{\boldsymbol{S}}_o$ are *unit vectors.*

We are projecting $\vec{\boldsymbol{a}}$, $\vec{\boldsymbol{b}}$, and $\vec{\boldsymbol{c}}$ on them.



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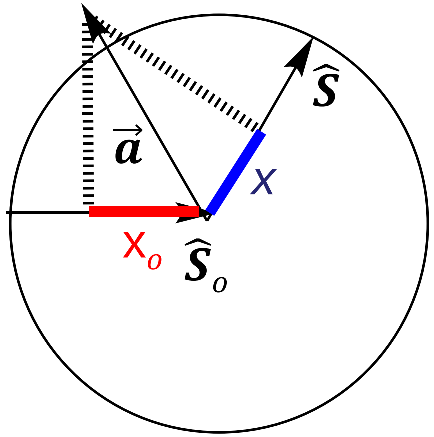
$\widehat{\boldsymbol{S}}$ and $\widehat{\boldsymbol{S}}_o$ are *unit vectors.*

For each axis, there are two projections we are interested in: x, and x_o.



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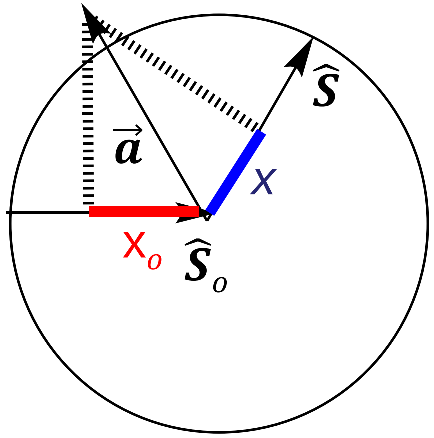
The sum of these two distances must be an integer multiple of λ, or else we get destructive interference.



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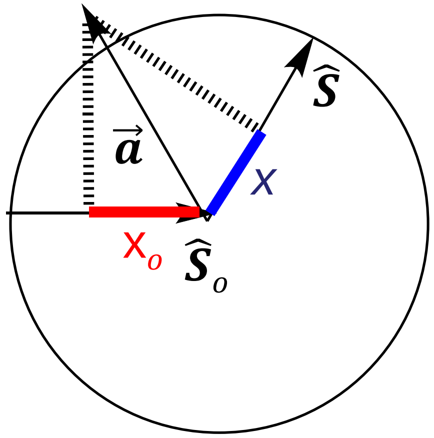
Where have we seen
this picture before?



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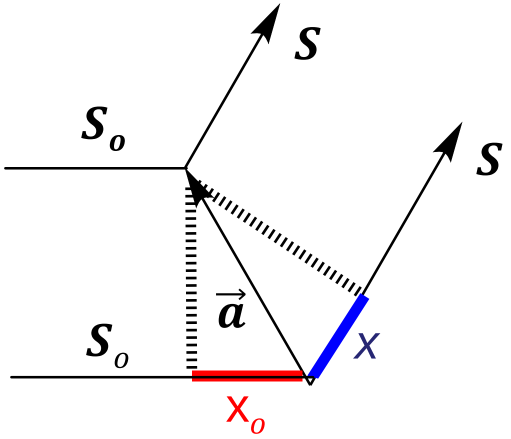

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$x = \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}}$

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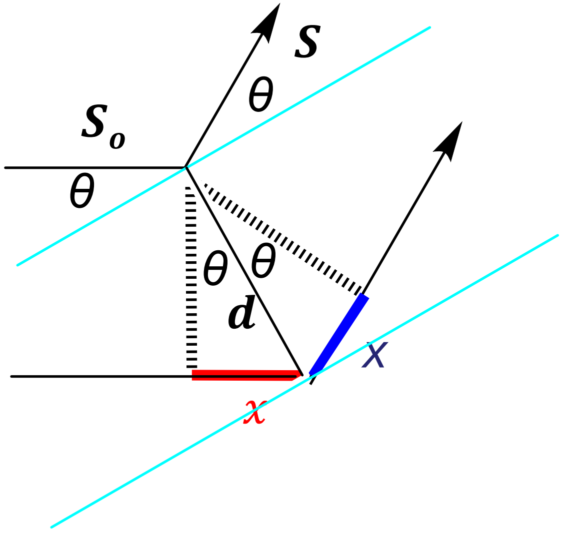
$2x = \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}} - \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}}_o = h\lambda$

$d = |\,\vec{\boldsymbol{a}}\,|$

$x = d \sin \theta$

Where have we seen this picture before?



2. Bragg

$$\vec{a} \cdot (\hat{S} - \hat{S}_o) = h\lambda$$
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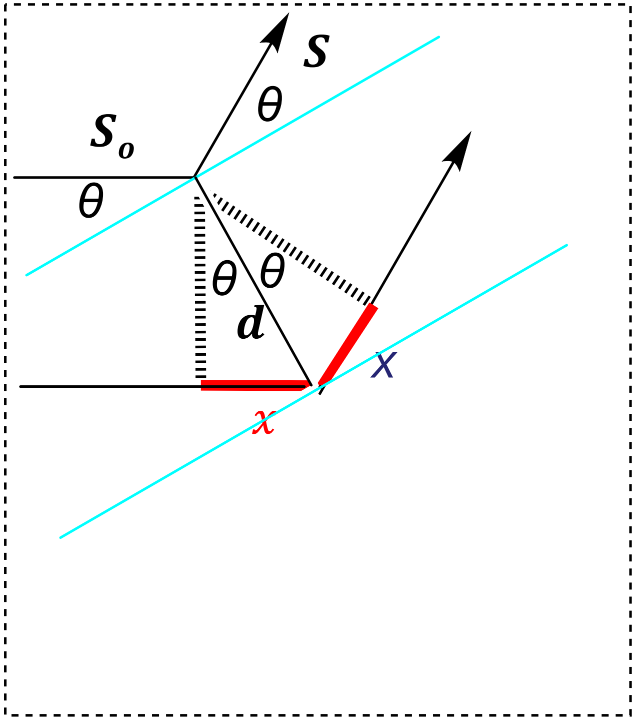
Bragg's law

$$2d \sin \theta = h\lambda$$



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$$x = \vec{a}\cdot\hat{S}$$
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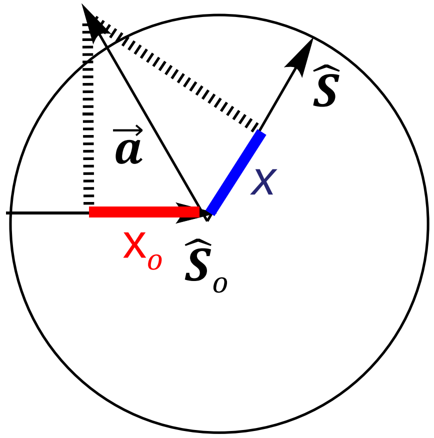
$$x + x_o = \vec{a}\cdot\hat{S} - \vec{a}\cdot\hat{S}_o = h\lambda$$

Projecting Bragg's law is just the special case when $\vec{a}, \hat{S}$, and $\hat{S}_o$ are all in the same plane.



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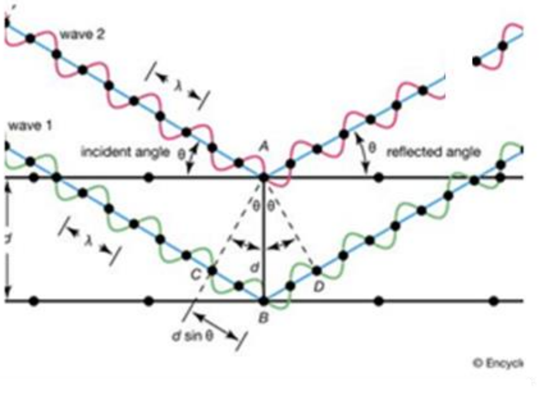

$$\vec{\boldsymbol{a}} \cdot \hat{\boldsymbol{S}} - \vec{\boldsymbol{a}} \cdot \hat{\boldsymbol{S}}_o = h\lambda$$

projecting $\hat{\boldsymbol{a}}$ onto $\hat{\boldsymbol{S}}$ and $\hat{\boldsymbol{S}}_o$

Beyond Bragg

$$\vec{a} \cdot (\hat{S} - \hat{S}_o) = h\lambda$$
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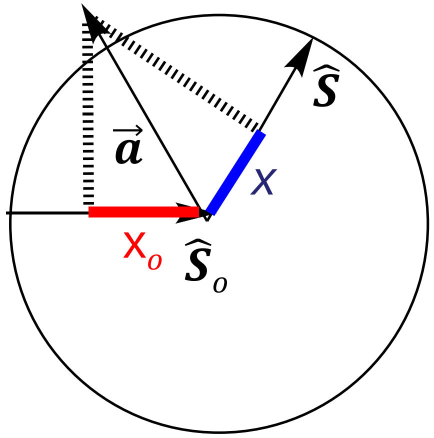
Real systems are more interesting. We are working with a *sphere,* not a *circle.* x and x_o do not have to be the same.



Beyond Bragg

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$$x = \vec{a}\cdot\hat{S}$$
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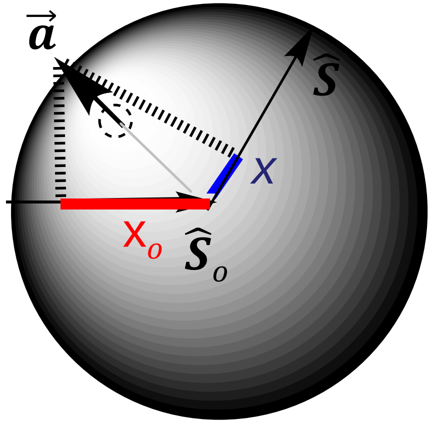
$$x + x_o = \vec{a}\cdot\hat{S} - \vec{a}\cdot\hat{S}_o = h\lambda$$

We could imagine the situation where $\vec{a}$ is out of the plane, and x and x_o, though their sum is the same, *are different values.*



Beyond Bragg

$$\vec{\boldsymbol{a}} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = h\lambda$$
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$$x = \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}}$$
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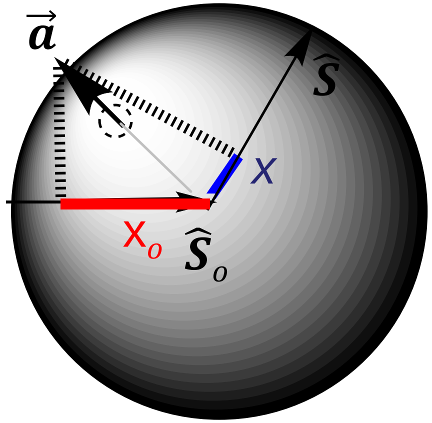
$$x + x_o = \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}} - \vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{S}}_o = h\lambda$$

This is inconvenient. How else can we visualize this?

How about projecting the other way – onto the unit vectors $\widehat{\boldsymbol{a}}, \widehat{\boldsymbol{b}}$, and $\widehat{\boldsymbol{c}}$ instead of onto $\widehat{\boldsymbol{S}}$ and $\widehat{\boldsymbol{S}}_o$?



Beyond Bragg

$$\vec{\boldsymbol{a}} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = h\lambda$$

$$\frac{\vec{\boldsymbol{a}}}{|\vec{\boldsymbol{a}}|} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = h\frac{\lambda}{|\vec{\boldsymbol{a}}|}$$

Beyond Bragg

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$$\widehat{\boldsymbol{a}} \cdot (\widehat{\boldsymbol{S}} - \widehat{\boldsymbol{S}}_o) = h\frac{\lambda}{a}$$

Beyond Bragg

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$$\widehat{\boldsymbol{S}} \cdot \widehat{\boldsymbol{a}} - \widehat{\boldsymbol{S}}_o \cdot \widehat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

All the vectors are unit vectors. Note that the right-hand side of the equation is unitless!

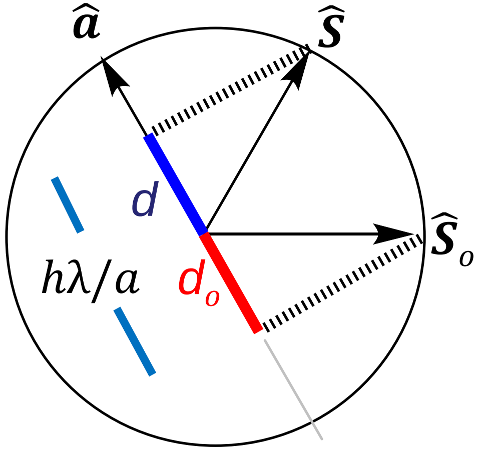
Beyond Bragg

$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{a}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$d = \hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{a}}$$
$$d_o = -\hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{a}}$$
$$d + d_o = h\lambda/a$$

Now both measurements are along the same line.

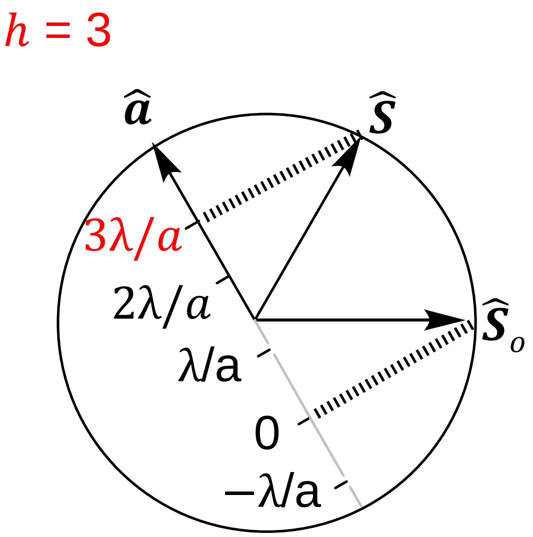
The sum is along a line, *even though the three vectors may not be in the same plane.*



Beyond Bragg

$$\widehat{\boldsymbol{S}} \cdot \widehat{\boldsymbol{a}} - \widehat{\boldsymbol{S}}_o \cdot \widehat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

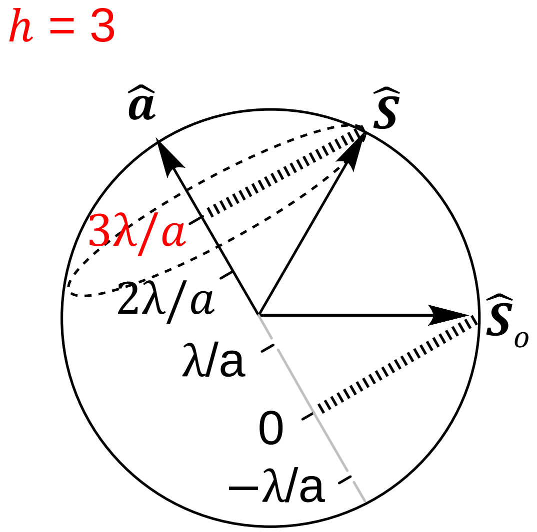
This suggests a number line. The Laue condition is satisfied at increments of λ/a along the ***a*** axis.



Beyond Bragg

$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{a}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

It's actually a *cone* of solutions, because we are dealing with a sphere, not just a circle.



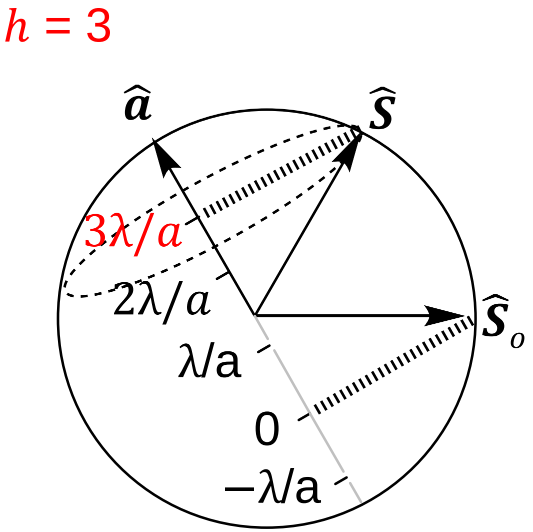
Beyond Bragg

$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{a}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{b}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

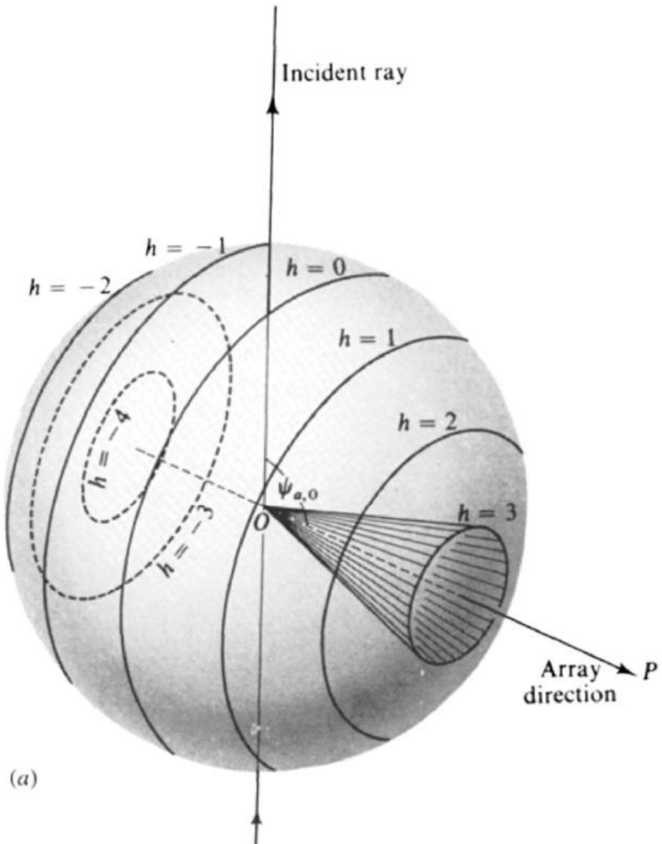
$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{c}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

… and this is true for the ***b*** and ***c*** axes, also.



3. Woolfson

The situation is visualized in Micheal Woolfson's book, *An Introduction to X-Ray Crystallography (1970)*



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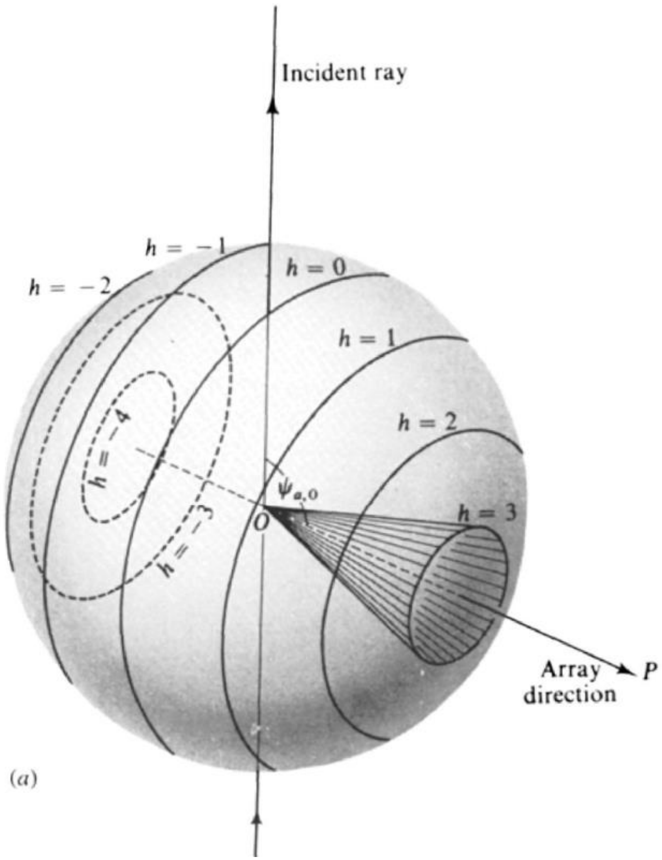
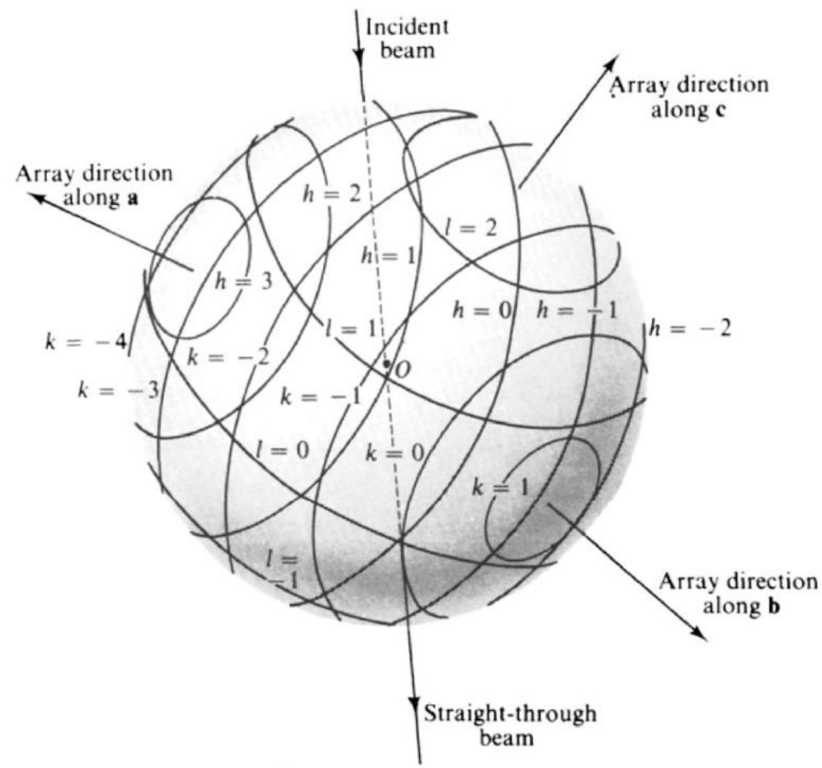
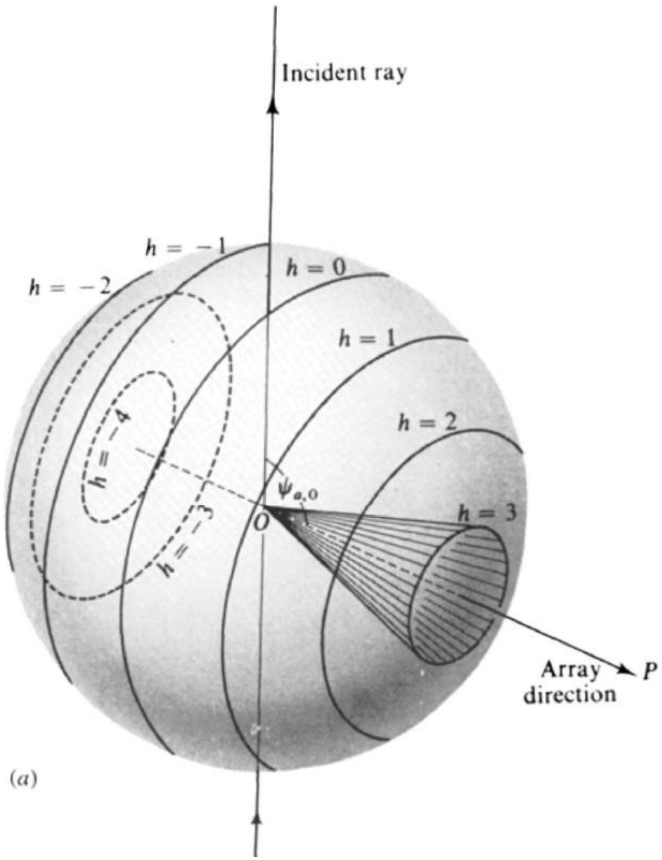


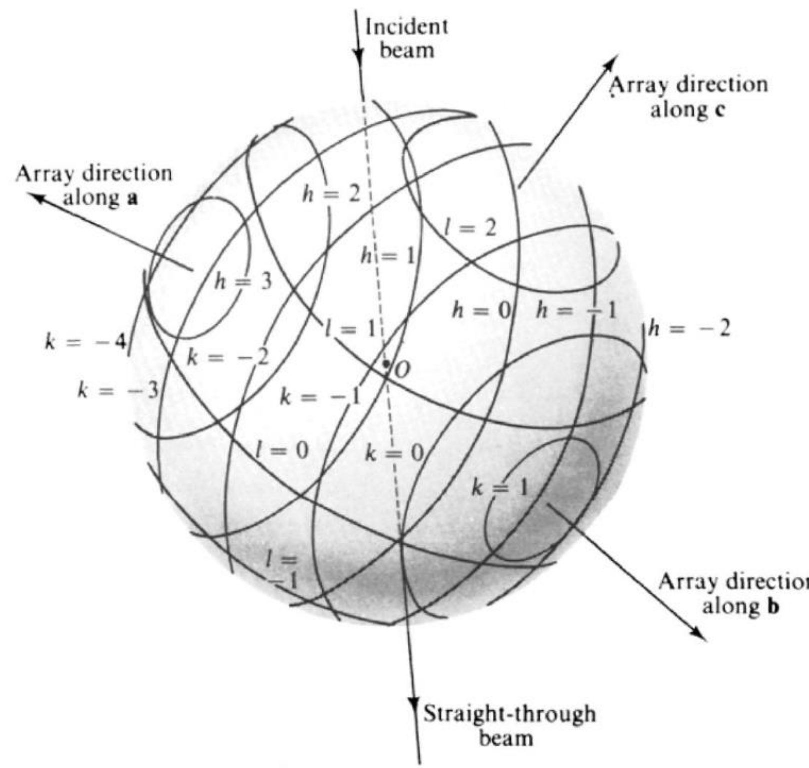


3. Woolfson

Can you find the place where all three Laue conditions are satisfied in this case? For ***a***, ***b***, and ***c***?



(a)



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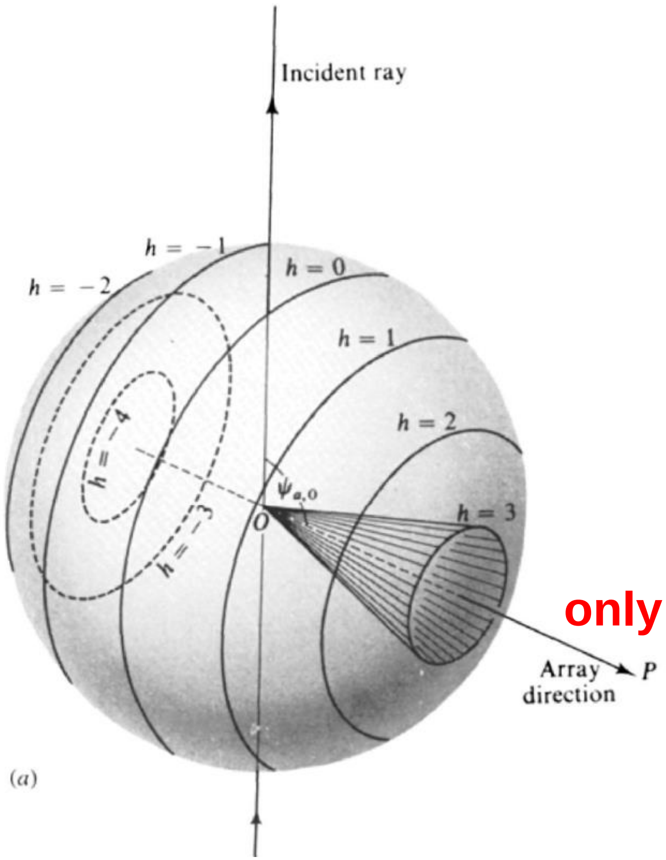
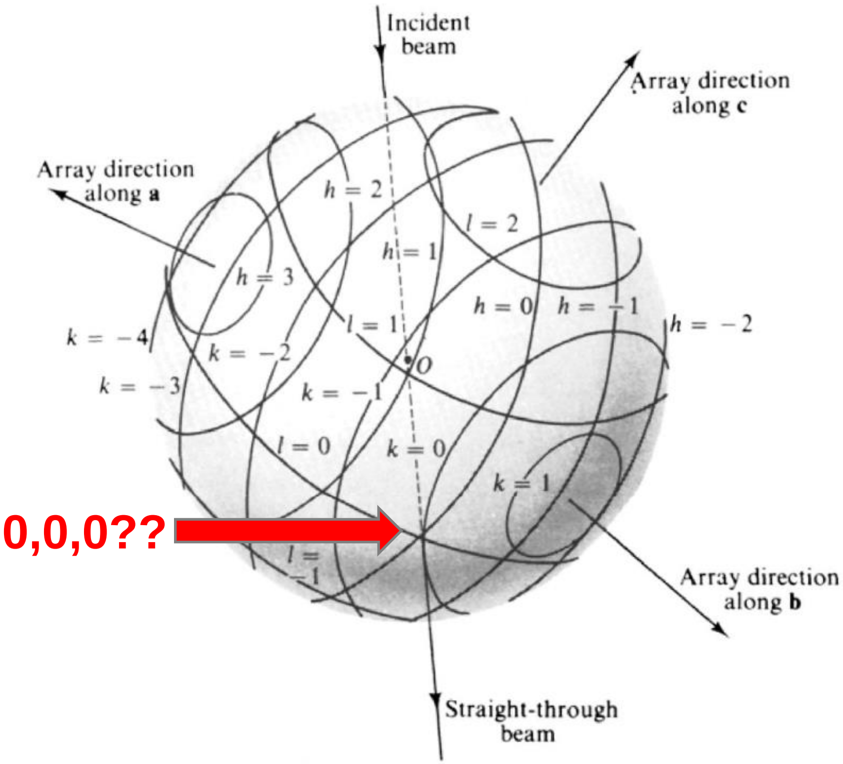


only 0,0,0??



3. Woolfson

$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{a}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

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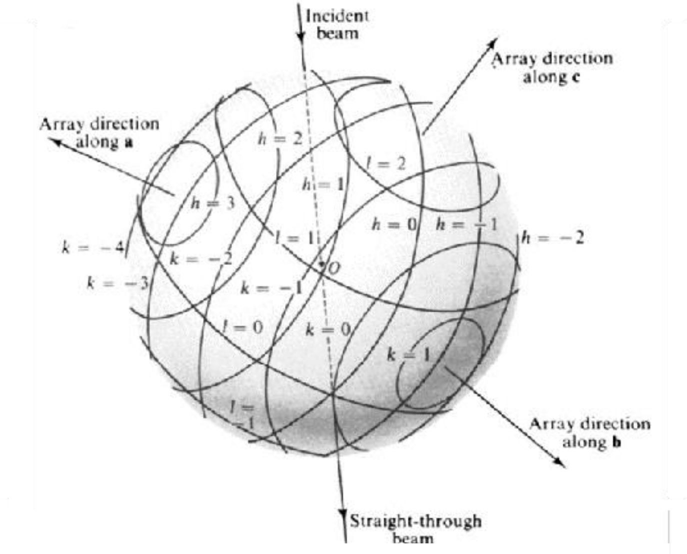
$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{c}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$



projecting $\hat{\boldsymbol{S}}$ and $\hat{\boldsymbol{S}}_o$ onto $\hat{\boldsymbol{a}}$, $\hat{\boldsymbol{b}}$, and $\hat{\boldsymbol{c}}$

Beyond Woolfson

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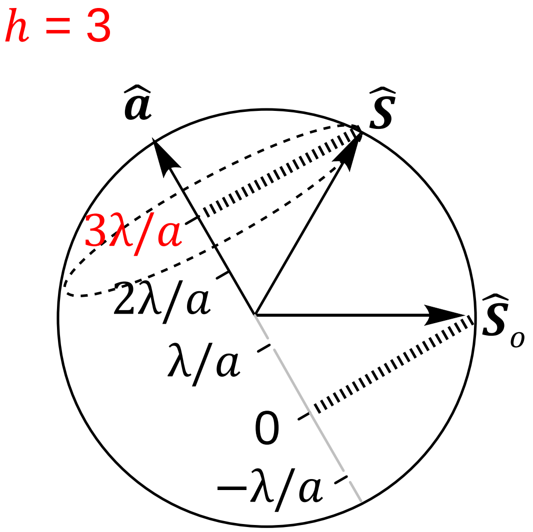
$$\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{c}} - \hat{\boldsymbol{S}}_o \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

Note that everything we are working with is a unit vector. We can regroup these another way.



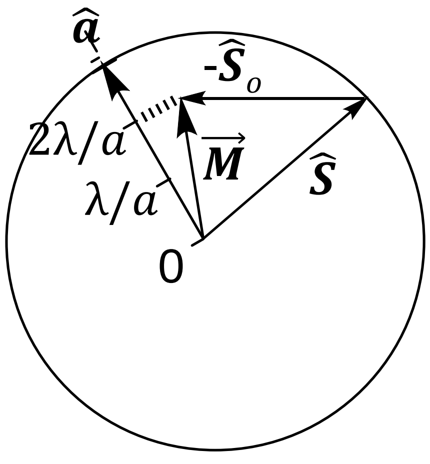
Beyond Woolfson

$$(\hat{\boldsymbol{S}} - \hat{\boldsymbol{S}}_o) \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$(\hat{\boldsymbol{S}} - \hat{\boldsymbol{S}}_o) \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

$$(\hat{\boldsymbol{S}} - \hat{\boldsymbol{S}}_o) \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

How about using the vector $\hat{\boldsymbol{S}} - \hat{\boldsymbol{S}}_o$ directly? Let's call this vector $\overrightarrow{\boldsymbol{M}}$. This is nice, because now our number scale starts at the origin.



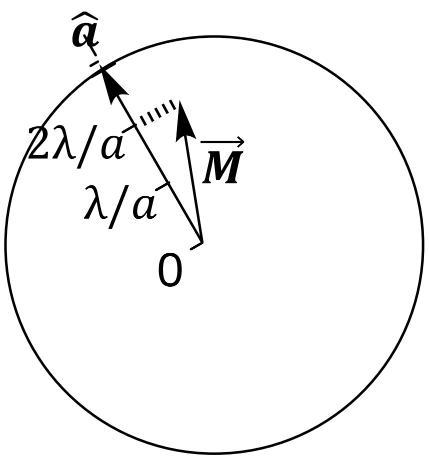
Beyond Woolfson

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

Note that $\overrightarrow{\boldsymbol{M}}$ is *fixed in space. There is only one* $\overrightarrow{\boldsymbol{M}}$ that can satisfy **all three** of these equations at the same time for a given set of $h\ k\ l$.



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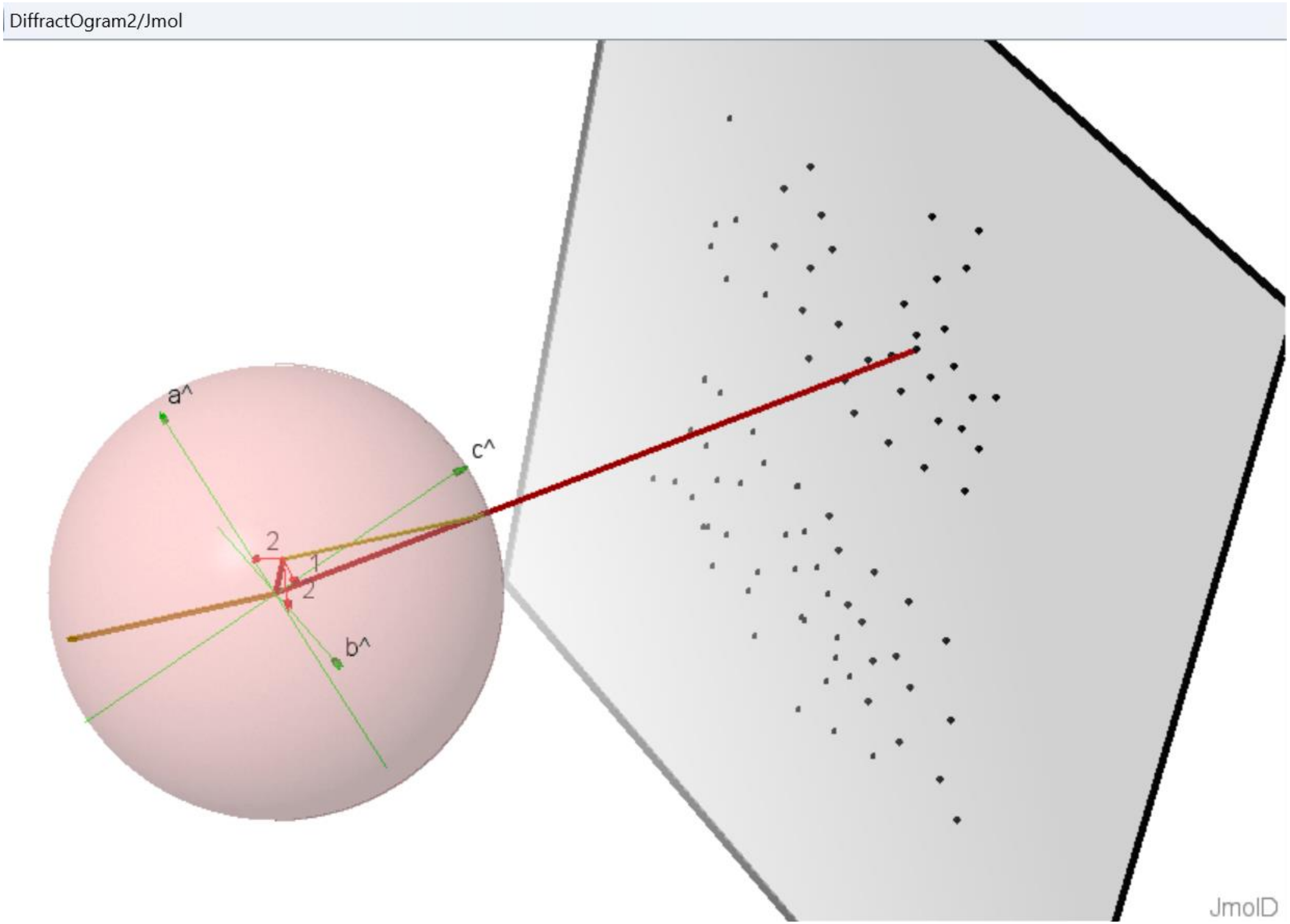
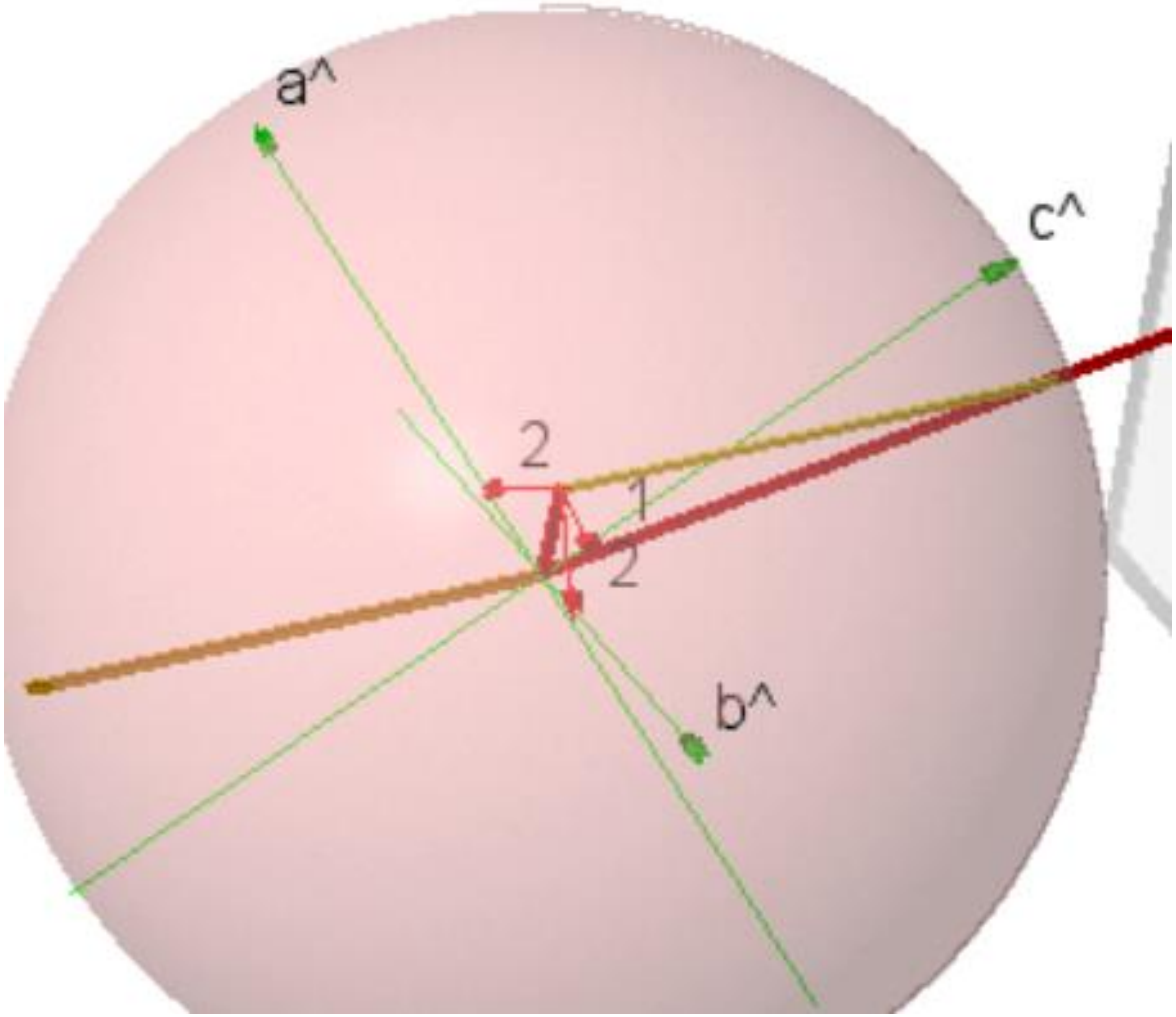
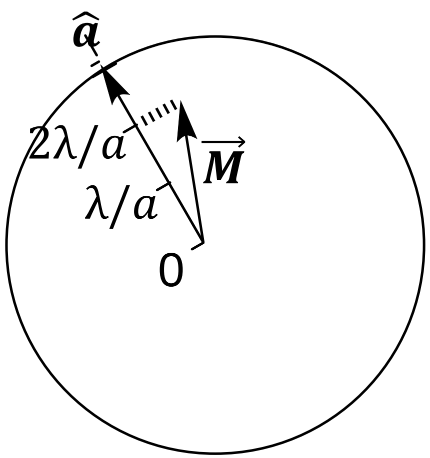


Beyond Woolfson





Beyond Woolfson

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

What else can we do
with projections?

Beyond Woolfson

$$\vec{\boldsymbol{M}} \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\vec{\boldsymbol{M}} \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

$$\vec{\boldsymbol{M}} \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

.

Let's switch the projection now: $\vec{\boldsymbol{a}}$ onto $\vec{\boldsymbol{M}}$ instead of $\vec{\boldsymbol{M}}$ onto $\vec{\boldsymbol{a}}$.

Beyond Woolfson

$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{a}}\, a = h\lambda$$
$$\overrightarrow{\boldsymbol{M}} \cdot \widehat{\boldsymbol{b}}\, b = k\lambda$$
$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{c}}\, \boldsymbol{c} = l\lambda$$

Let’s switch the projection now: $\vec{\boldsymbol{a}}$ onto $\overrightarrow{\boldsymbol{M}}$ instead of $\overrightarrow{\boldsymbol{M}}$ onto $\vec{\boldsymbol{a}}$.

Beyond Woolfson

$$\overrightarrow{\boldsymbol{M}} \cdot \vec{\boldsymbol{a}} = h\lambda$$
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Let's switch the projection now: $\vec{\boldsymbol{a}}$ onto $\overrightarrow{\boldsymbol{M}}$ instead of $\overrightarrow{\boldsymbol{M}}$ onto $\vec{\boldsymbol{a}}$.

Beyond Woolfson

$$\vec{\boldsymbol{a}} \cdot \overrightarrow{\boldsymbol{M}} = h\lambda$$
$$\vec{\boldsymbol{b}} \cdot \overrightarrow{\boldsymbol{M}} = k\lambda$$
$$\vec{\boldsymbol{c}} \cdot \overrightarrow{\boldsymbol{M}} = l\lambda$$

And now we need to unitize $\overrightarrow{\boldsymbol{M}}$ by dividing both sides by $|\overrightarrow{\boldsymbol{M}}|$.

Beyond Woolfson

$$\vec{\boldsymbol{a}} \cdot \frac{\overrightarrow{\boldsymbol{M}}}{|\overrightarrow{\boldsymbol{M}}|} = h\lambda / |\overrightarrow{\boldsymbol{M}}|$$

$$\vec{\boldsymbol{b}} \cdot \frac{\overrightarrow{\boldsymbol{M}}}{|\overrightarrow{\boldsymbol{M}}|} = k\lambda / |\overrightarrow{\boldsymbol{M}}|$$

$$\vec{\boldsymbol{c}} \cdot \frac{\overrightarrow{\boldsymbol{M}}}{|\overrightarrow{\boldsymbol{M}}|} = l\lambda / |\overrightarrow{\boldsymbol{M}}|$$

…and since $\frac{\overrightarrow{\boldsymbol{M}}}{|\overrightarrow{\boldsymbol{M}}|}$ is $\widehat{M}$…

Beyond Woolfson

$$\vec{\boldsymbol{a}} \cdot \widehat{\boldsymbol{M}} = h\lambda/|\overrightarrow{\boldsymbol{M}}|$$

$$\vec{\boldsymbol{b}} \cdot \widehat{\boldsymbol{M}} = k\lambda/|\overrightarrow{\boldsymbol{M}}|$$

$$\vec{\boldsymbol{c}} \cdot \widehat{\boldsymbol{M}} = l\lambda/|\overrightarrow{\boldsymbol{M}}|$$

Almost there!

…moving h, k, and l over to the other side of the equation…

Beyond Woolfson

$$\frac{\vec{\boldsymbol{a}}}{h} \cdot \widehat{\boldsymbol{M}} = \lambda / |\overrightarrow{\boldsymbol{M}}|$$

$$\frac{\vec{\boldsymbol{b}}}{k} \cdot \widehat{\boldsymbol{M}} = \lambda / |\overrightarrow{\boldsymbol{M}}|$$

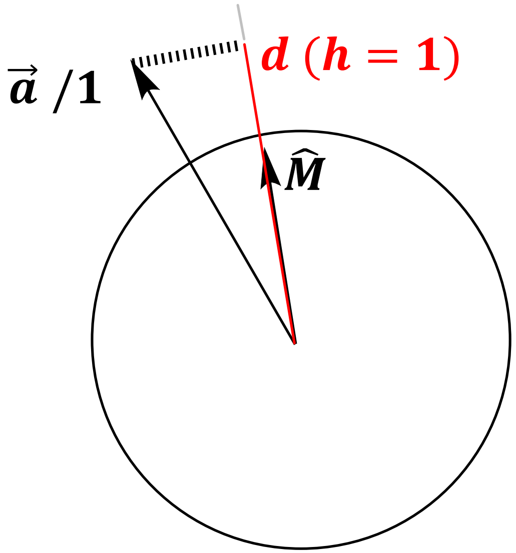
$$\frac{\vec{\boldsymbol{c}}}{l} \cdot \widehat{\boldsymbol{M}} = \lambda / |\overrightarrow{\boldsymbol{M}}|$$

…and let's just call the quantity $\lambda / |\overrightarrow{\boldsymbol{M}}|$ "$\boldsymbol{d}$“…

$$\frac{\vec{\boldsymbol{a}}}{h} \cdot \widehat{\boldsymbol{M}} = \frac{\vec{\boldsymbol{b}}}{k} \cdot \widehat{\boldsymbol{M}} = \frac{\vec{\boldsymbol{c}}}{l} \cdot \widehat{\boldsymbol{M}} = d$$

Projecting $\vec{\boldsymbol{a}}/_{h}$, $\vec{\boldsymbol{b}}/_{k}$, or $\vec{\boldsymbol{c}}/_{l}$ onto $\widehat{\boldsymbol{M}}$ all give the same value.

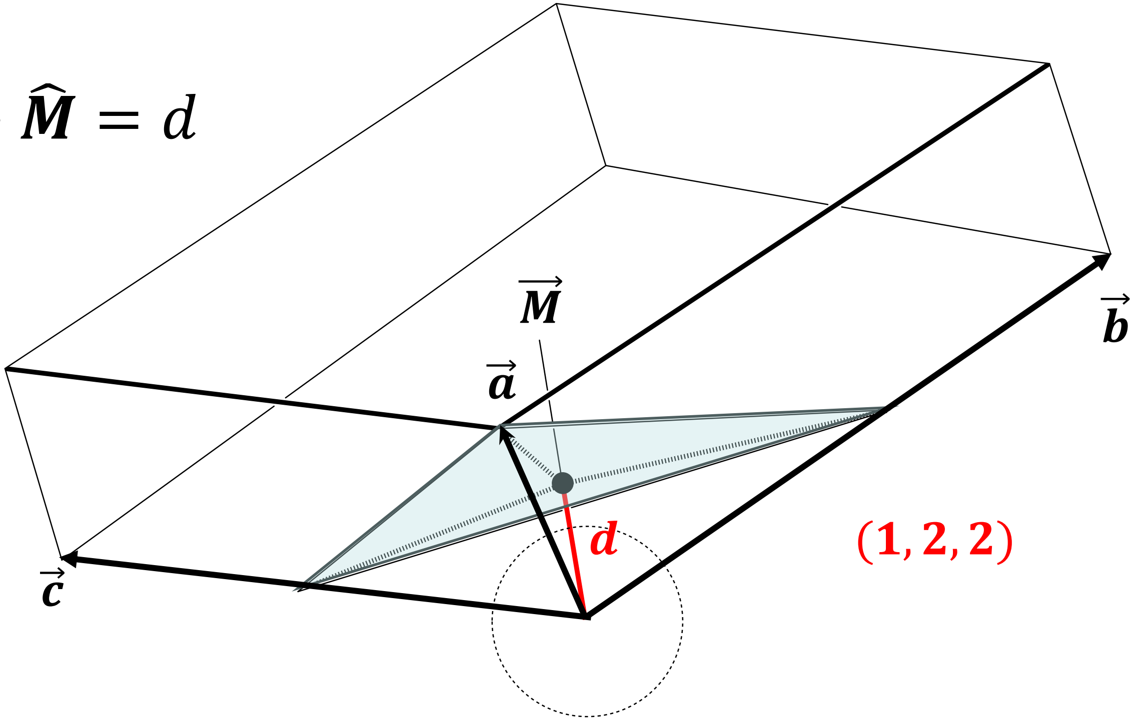
How cool is that???



4. Miller

$$\frac{\vec{\boldsymbol{a}}}{h} \cdot \widehat{\boldsymbol{M}} = \frac{\vec{\boldsymbol{b}}}{k} \cdot \widehat{\boldsymbol{M}} = \frac{\vec{\boldsymbol{c}}}{l} \cdot \widehat{\boldsymbol{M}} = d$$

.

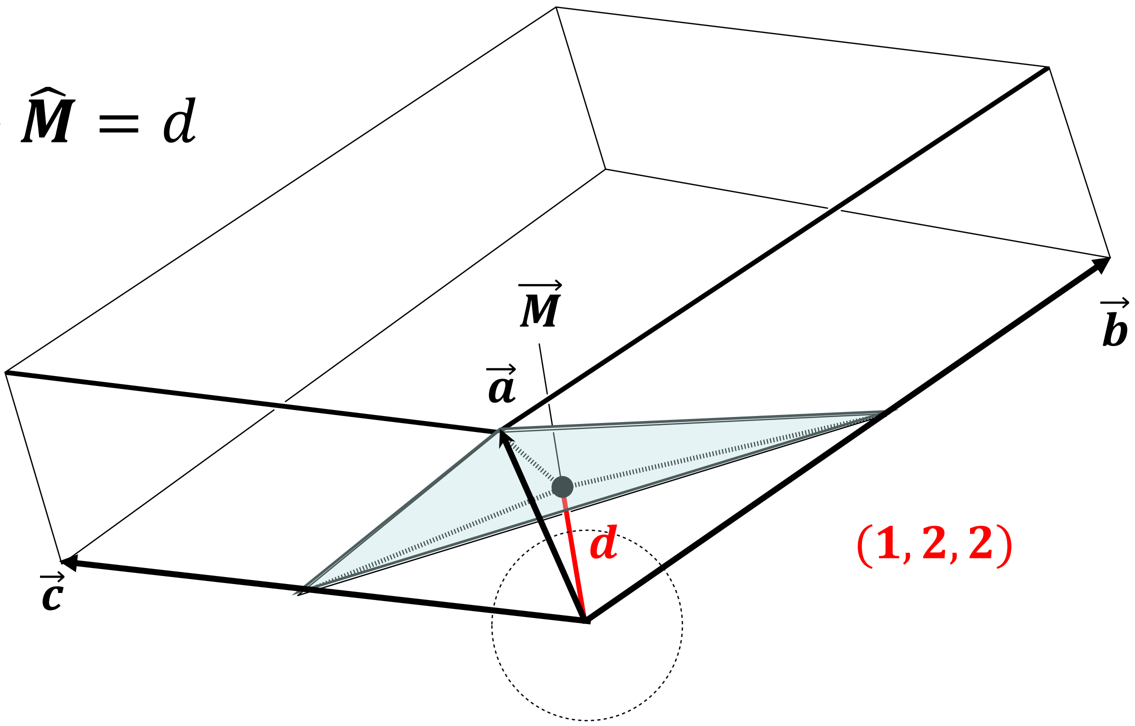


This is the origin of the idea of a Miller plane. The three fractional axis points define a plane that is the distance d from the origin.

4. Miller

$$\frac{\vec{\boldsymbol{a}}}{h} \cdot \widehat{\boldsymbol{M}} = \frac{\vec{\boldsymbol{b}}}{k} \cdot \widehat{\boldsymbol{M}} = \frac{\vec{\boldsymbol{c}}}{l} \cdot \widehat{\boldsymbol{M}} = d$$

.



projecting $\vec{\boldsymbol{a}}/\boldsymbol{h}$, $\vec{\boldsymbol{b}}/\boldsymbol{k}$, and $\vec{\boldsymbol{c}}/\boldsymbol{l}$ onto $\widehat{\boldsymbol{M}}$

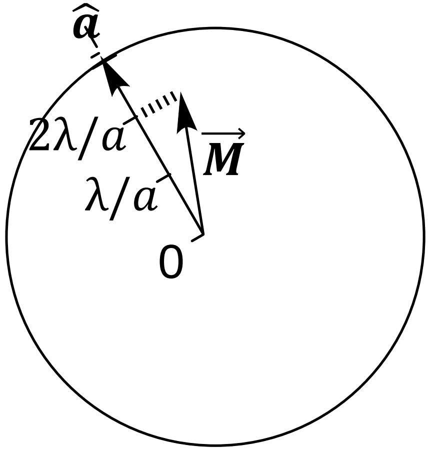
Beyond Miller

$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

Are we done? Almost. What about the reciprocal lattice?



Beyond Miller

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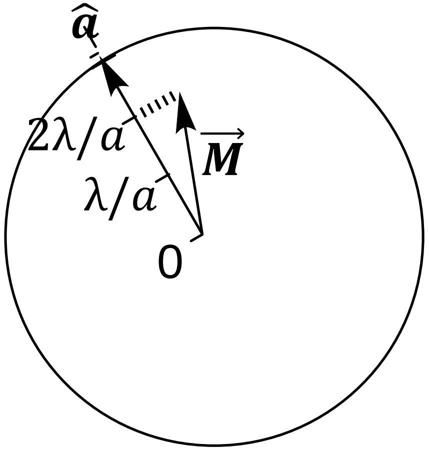
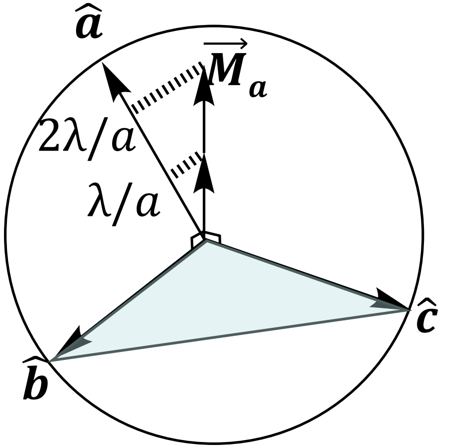
The idea is to construct a basis for $\overrightarrow{\boldsymbol{M}}$ that has the coordinates (h, k, l).



Beyond Miller

$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$
$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$
$$\overrightarrow{\boldsymbol{M}} \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

This is achieved by using the specific unit vectors $\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}}$, $\overrightarrow{\boldsymbol{M}}_{\boldsymbol{b}}$, and $\overrightarrow{\boldsymbol{M}}_{\boldsymbol{c}}$ that are perpendicular to the *bc, ca,* and *ab* planes, respectively.



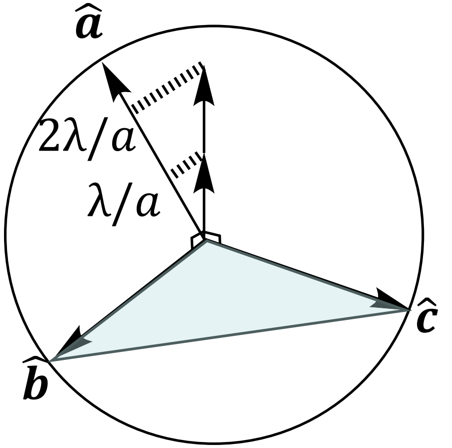
Beyond Miller

$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} \cdot \widehat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} \cdot \widehat{\boldsymbol{b}} = 0$$

$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} \cdot \widehat{\boldsymbol{c}} = 0$$

$\widehat{\boldsymbol{M}}_{\boldsymbol{a}}$ “selects” for the ***a*** axis.

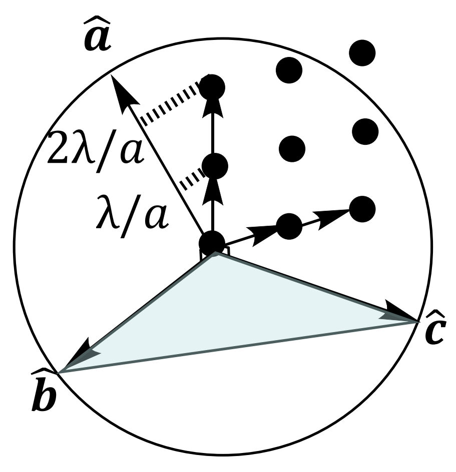


$$\vec{\boldsymbol{M}}_a \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a} \qquad \vec{\boldsymbol{M}}_b \cdot \hat{\boldsymbol{a}} = \mathbf{0} \qquad \vec{\boldsymbol{M}}_c \cdot \hat{\boldsymbol{a}} = \mathbf{0}$$

$$\vec{\boldsymbol{M}}_a \cdot \hat{\boldsymbol{b}} = 0 \qquad \vec{\boldsymbol{M}}_b \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b} \qquad \vec{\boldsymbol{M}}_c \cdot \hat{\boldsymbol{b}} = 0$$

$$\vec{\boldsymbol{M}}_a \cdot \hat{\boldsymbol{c}} = 0 \qquad \vec{\boldsymbol{M}}_b \cdot \hat{\boldsymbol{c}} = 0 \qquad \vec{\boldsymbol{M}}_c \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

The collection of these vectors forms a discrete lattice. In this case, it is a *unitless* lattice. But it is essentially the *reciprocal lattice.* It's just unitless.



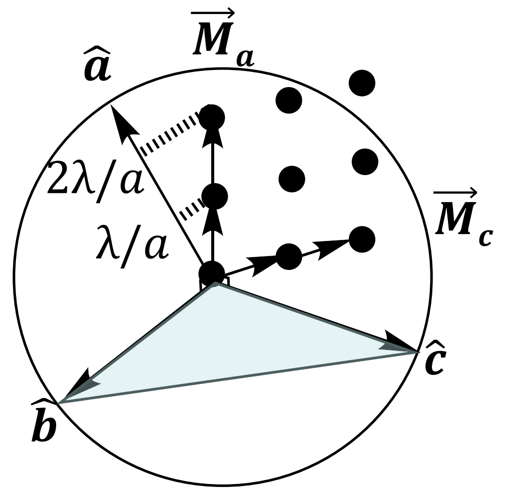
5. Gibbs

$$\vec{\boldsymbol{M}}_{\boldsymbol{a}} \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$

$$\vec{\boldsymbol{M}}_{\boldsymbol{b}} \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$

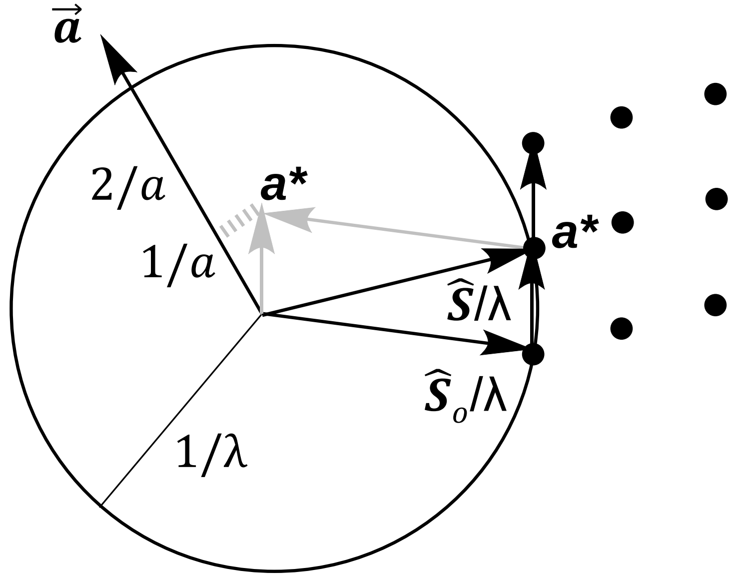
$$\vec{\boldsymbol{M}}_{\boldsymbol{c}} \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

Willard Gibbs published the general idea of a reciprocal lattice in 1901.

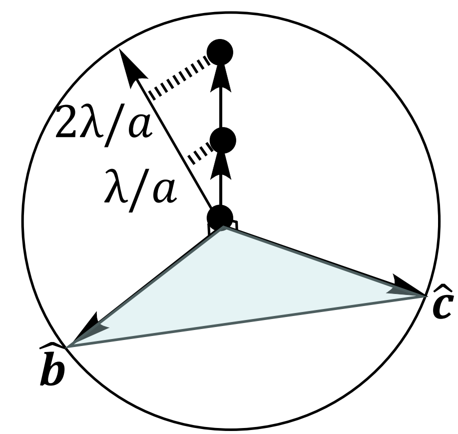


https://www.google.com/books/edition/Vector_Analysis/j6QKAAAAYAAJ (p 81)

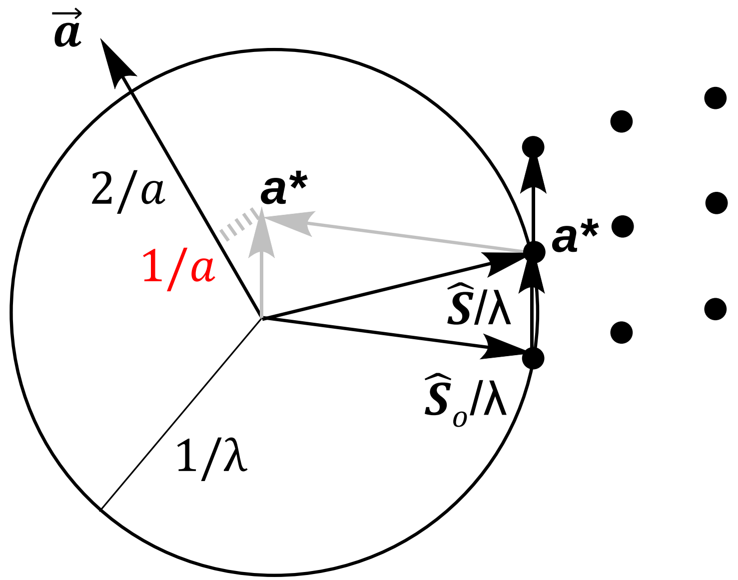
6. Ewald



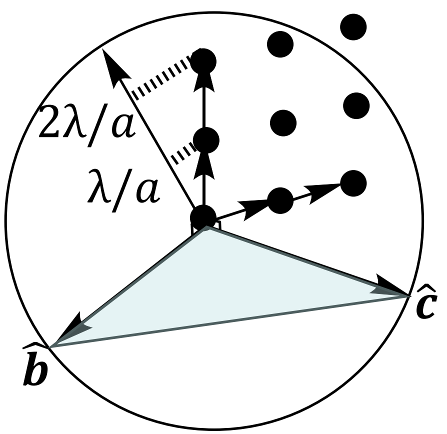
Ewald thought of this somewhat differently. His sphere has a radius of $\frac{1}{\lambda}$, And his lattice is scaled by $\frac{1}{\lambda}$ and moved over to have its origin at the end of the $\widehat{\boldsymbol{S}}_o/\lambda$ vector. How curious!!



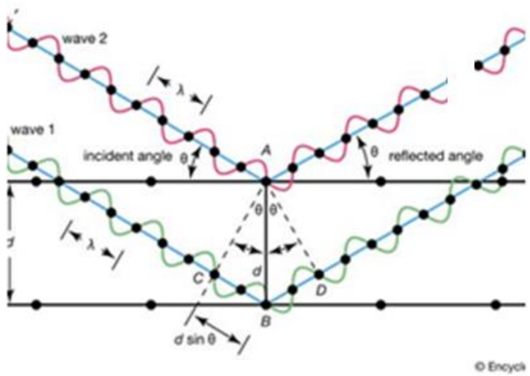
6. Ewald



Note that the distances between lattice points along the ***a**** axis is *not* $1/a$. That value is the projected distance along the ***a*** axis. The distance between lattice points is larger and not particularly relevant.



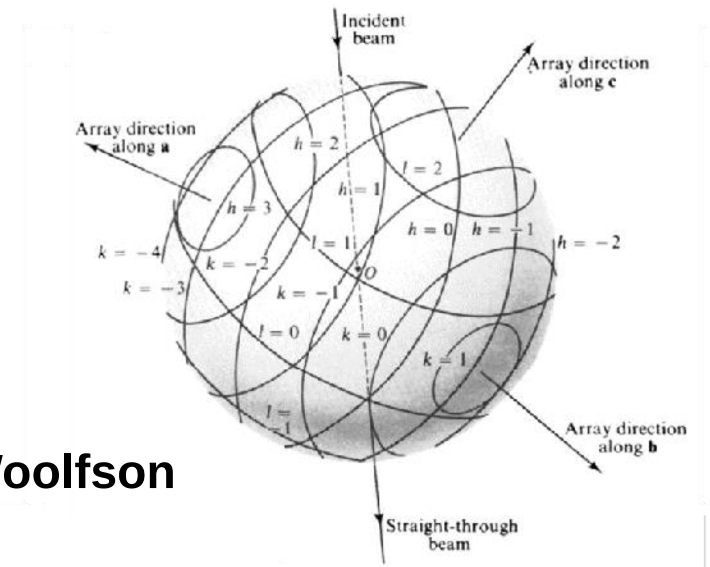
Different visualizations; same eqn



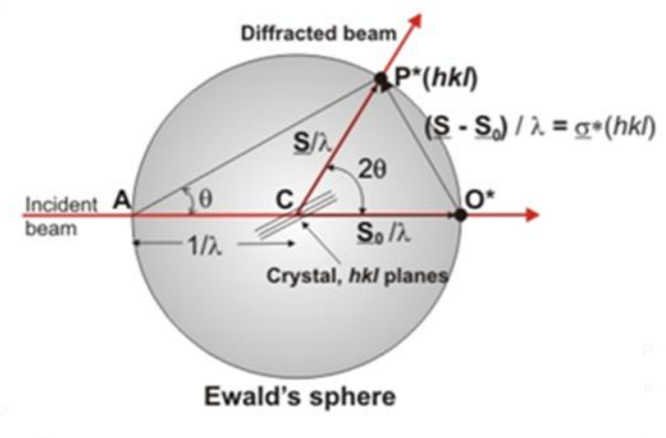
Bragg

Laue

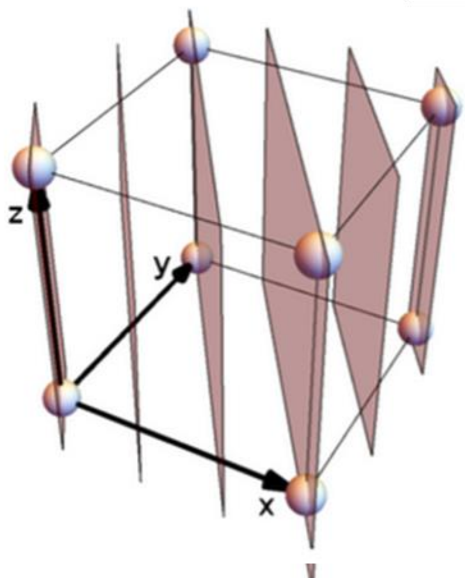
$$\mathbf{a}\cdot\mathbf{s} = h$$
$$\mathbf{b}\cdot\mathbf{s} = k$$
$$\mathbf{c}\cdot\mathbf{s} = l$$



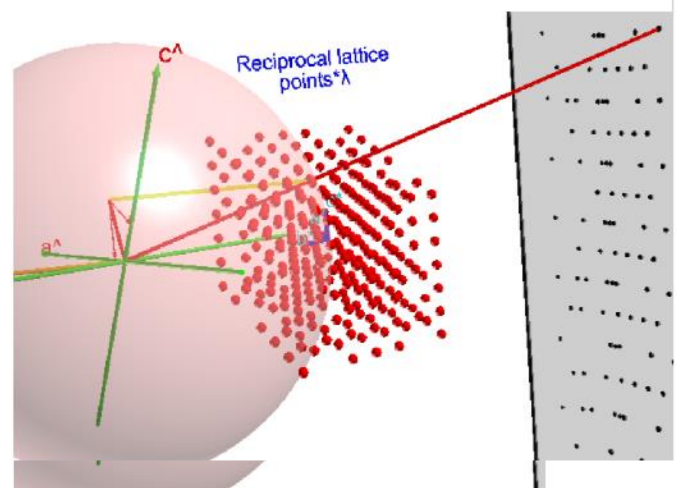
Woolfson



Ewald

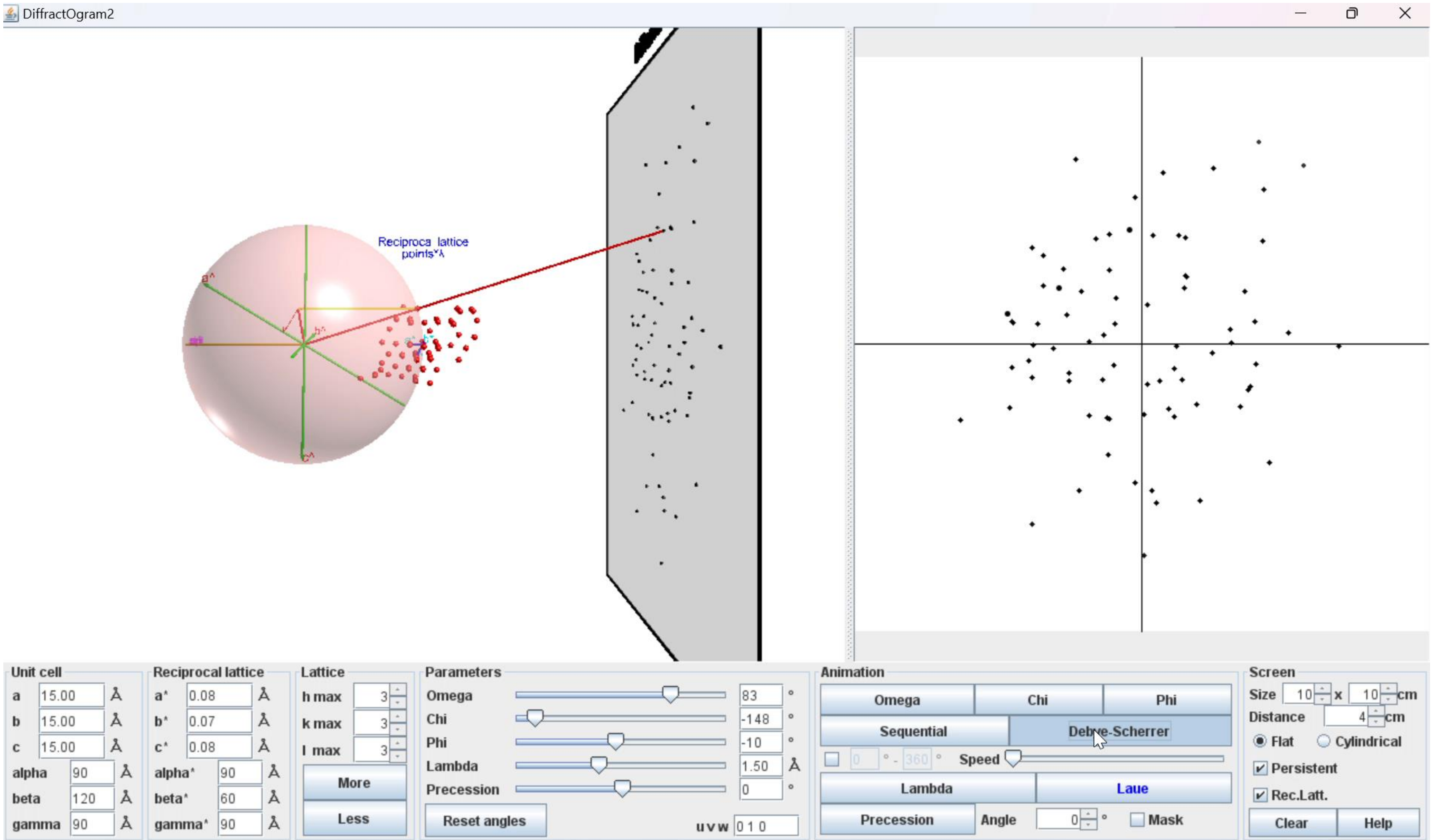


Miller



Gibbs

Loads of fun!



Thank you!

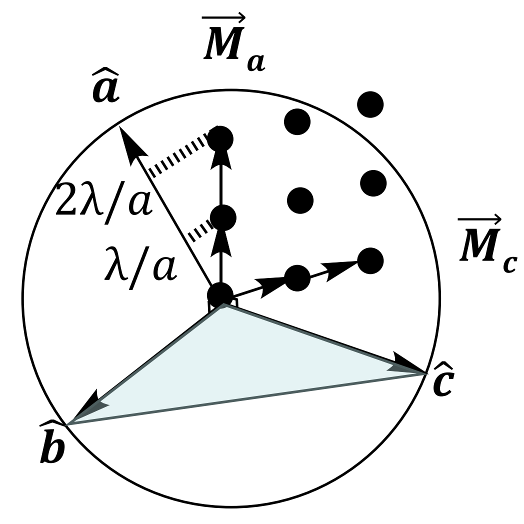
ps. What exactly is the reciprocal lattice?

$$\vec{\boldsymbol{M}} \cdot \hat{\boldsymbol{a}} = h\frac{\lambda}{a}$$
$$\vec{\boldsymbol{M}} \cdot \hat{\boldsymbol{b}} = k\frac{\lambda}{b}$$
$$\vec{\boldsymbol{M}} \cdot \hat{\boldsymbol{c}} = l\frac{\lambda}{c}$$

or

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{M}}_{\boldsymbol{a}} = h\lambda$$
$$\vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{M}}_{\boldsymbol{b}} = k\lambda$$
$$\vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{M}}_{\boldsymbol{c}} = l\lambda$$

We need the specific unit vectors $\vec{\boldsymbol{M}}_{\boldsymbol{a}}$, $\vec{\boldsymbol{M}}_{\boldsymbol{b}}$, and $\vec{\boldsymbol{M}}_{\boldsymbol{c}}$ that are perpendicular to the *bc, ca,* and *ab* planes, respectively.



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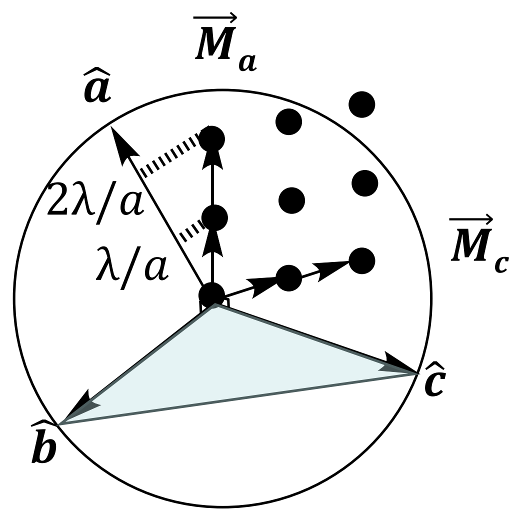
The way this is done is to use cross products. $\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}$ is a vector perpendicular to both $\vec{\boldsymbol{b}}$ and $\vec{\boldsymbol{c}}$. All we need to do is scale that to be what we need on the right.



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$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} = h\lambda\left(\frac{\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}{\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}\right)$$

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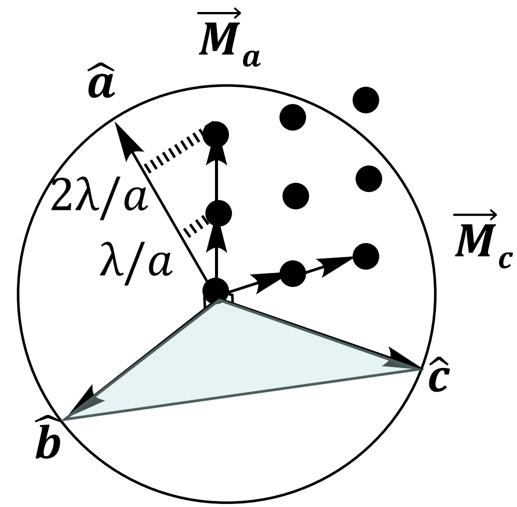

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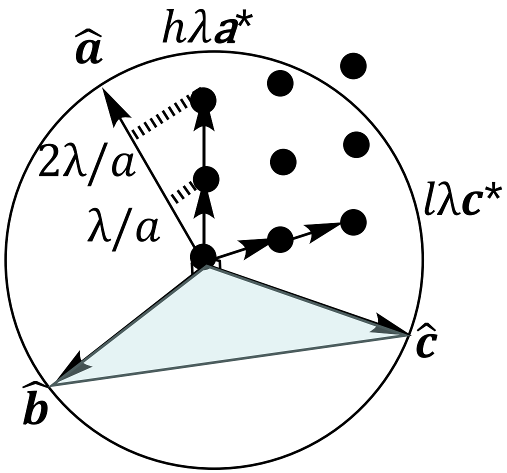
$$\vec{\boldsymbol{c}} \cdot \overrightarrow{\boldsymbol{M}}_{\boldsymbol{c}} = l\lambda$$

$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} = h\lambda\left(\frac{\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}{\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}\right)$$
$$= h\lambda\boldsymbol{a}^*$$

The usual way this is expressed is to use just the quantity in parentheses for the "reciprocal lattice" vector.



ps. What exactly is the reciprocal lattice?

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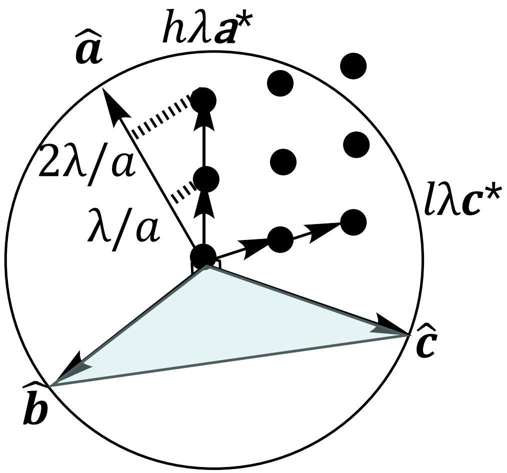
$$\vec{\boldsymbol{c}} \cdot \overrightarrow{\boldsymbol{M}}_{\boldsymbol{c}} = l\lambda$$

$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} = h * \lambda \left(\frac{\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}{\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}} \right)$$

I like to include λ because it reminds us that it is all about getting integral multiples of the X-ray wavelength, and it allows us to use the unit circle for all the visualizations.



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$$\overrightarrow{\boldsymbol{M}}_{\boldsymbol{a}} = h * \lambda\left(\frac{\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}{\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}}\right)$$

And it turns the “reciprocal” lattice into a *selector* lattice. Selecting out the specific independent components that all must satisfy the Bragg equation.